1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist synthesis, some MIBC tumors with high mutational signals (TMB/bTMB/HRD) respond to checkpoint blockade while others with comparable mutational metrics do not; why does genomic immunogenicity sometimes fail to translate into effective anti-tumor immunity? 2) Tier 1 paths used: I built on (a) the TMB → immunotherapy response path (establishes mutation-derived neoantigen load as a driver of response), (b) the HRD score path (links DNA-repair deficiency/TP53-driven genomic instability to neoantigen quality and load), and (c) the bTMB path (provides a blood-based surrogate for tissue mutational signals). These converge on a mechanistic question about whether DNA-repair–driven neoantigens enable organized local immune structures that permit response and whether bTMB captures that biology. 3) Tier 2 evidence supporting the claim: TP53 and genomic instability provide a mechanistic source of HRD-associated neoantigens; PD-L1 and MSI are additional immune-context modulators that could influence whether an antigen-rich tumor elicits TLS formation and productive immunity; liquid biopsy feasibility supports using bTMB for non-invasive stratification. 4) Why I chose therapy_response as the endpoint: The core paths center on predicting and monitoring checkpoint inhibitor responsiveness (immunotherapy), so a therapy_response proxy is the most direct, clinically actionable endpoint for validating predictive performance and mechanism. 5) Deliberate exclusions: I excluded exploratory/metabolic and poorly linked paths (GLU/urine biomarkers, pZVR, broad prognostic markers) because they lack clear mechanistic connection to immune priming/TLS formation in the TMB–HRD axis; I also limited anchors to HRD and bTMB to keep the hypothesis sharp and mechanistic (TMB, TP53, PD-L1 moved to secondary_modifiers)?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC), TLS-like lymphoid aggregates enable effective antitumor immunity and predict favorable therapy_response to checkpoint inhibitors only when tumor neoantigen quality arises from homologous recombination deficiency (high HRD_score). Specifically, TLS-positive tumors with high HRD_score will show higher therapy_response rates than TLS-positive tumors with low HRD_score or TLS-negative tumors, due to HRD-driven clonal neoantigens that sustain intratumoral antigen presentation. Conversely, tumors with high bTMB but low HRD_score will have abundant circulating mutations but insufficient TLS-driven immune organization, resulting in lower therapy_response despite high bTMB. This framework predicts that TLS presence and HRD_score interact to modulate immunotherapy responsiveness, with PD-L1 and TP53-associated genomic instability acting as modulators of this axis; bTMB serves as a blood-based surrogate that may reflect the mutational burden but does not substitute for HRD-driven neoantigen quality in predicting TLS-mediated responses.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	HRD_score, bTMB
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

HRD_score: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

bTMB: bTMB/HRD) respond to checkpoint blockade while others with comparable mutational metrics do not; why does genomic immunogenicity sometimes fail to translate into effective anti-tumor immunity? 2) Tier 1 paths used: I built on (a) the TMB → immunotherapy response path (establishes mutation-derived neoantigen load as a driver of response), (b) the HRD score path (links DNA-repair deficiency/TP53-driven genomic instability to neoantigen quality and load), and (c) the bTMB path (provides a blood-based surrogate for tissue mutational signals).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	PD-L1	Supporting modifier from knowledge graph
Secondary	TMB	Supporting modifier from knowledge graph
Secondary	MSI	Supporting modifier from knowledge graph
Secondary	genomic_instability	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective analyses of MIBC cohorts treated with checkpoint inhibitors with available tissue for TLS assessment (IHC for TLS components, e.g., CD3, CD20, CXCL13) and sequencing to derive HRD_score and TMB;

Prediction 2: obtain bTMB from cfDNA when tissue is unavailable;

Prediction 3: multivariate modeling of therapy_response with TLS status, HRD_score, and bTMB, including interaction terms;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PD-L1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.